

more opaque as larger. The clear mapping on the left is preferred, and the unclear mapping on the right must be rescued with adjustments to color combinations (Schloss et al., 2018).

One could imagine many alternative rules that could govern the mappings from numbers to visual properties. A computer vision system could be programmed to map larger numbers to higher positions, as humans do, or just as easily programmed to map them to lower positions, leftward positions, shorter lengths, or lighter values. Continua need not even be mapped consistently over continuous space: one could map the number 1 to position 1, 2 to 9, 3 to 5, and so on; as long as that translation is available to the computer vision system, it could easily recover the original values. The fixed architecture of the human visual system could not efficiently process those mappings.

Respect associations between visualization designs and data types

We have so far focused only on *metric* visualization—the visualization of continuous magnitudes that can be positive or negative. There are other types of data (Stevens, 1946) that are frequently shown in visualizations (Bertin, 1983; Mackinlay, 1986; Munzner, 2014). One is nominal data, categorical values that can be counted or can be used to divide a set of metric numbers. Data sets often include both metric and nominal data. For example, a company might have a spreadsheet recording sales with columns of metric values (number of orders, sales in dollars, shipping time in hours) and columns of nominal values (customer name, product ID, country code). One might sum over all of the metric sales numbers or split those numbers by product ID to compute averages for each ID. Another type of data falls in between metric and nominal: *ordinal* values, which represent ranks that can be compared and ordered but do not reflect continuous metric differences. Examples include stars or dollar signs for restaurant ratings or price levels; designations of coach, business, and first class on an airplane; or rankings of colleges in a magazine. These values may have been generated with metric data (e.g., the cost of airplane tickets, numeric ratings of colleges), but they do not contain metric information per se.

Viewers expect particular types of data to be mapped to particular visual properties. Metric data are typically mapped to position, length, area, angle, or intensity. Nominal data can be mapped to position (e.g., a bar graph showing sales for each product ID might spread those IDs from left to right or from top to bottom) or to color or shape (e.g., to identify each point's country code on a scatterplot pitting sales against shipping

time). But if nominal data are plotted with lengths or areas, viewers can become confused. An example of this type of confusing mapping is depicted in Figure 18 (adapted from Mackinlay, 1986). The varied lengths of the bars make their values “feel” metric, such that the viewer wonders why Japan is “larger” than Italy.

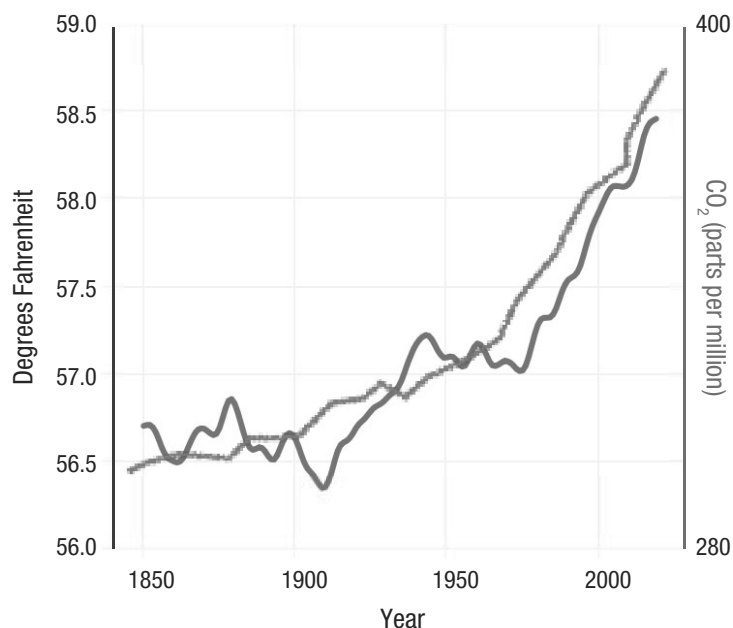
Figure 18 also shows that the visual system responds to points or bars as separate objects, whereas a line connecting points “feel like” a single object that has been stretched or has moved over time and left a path (Tversky, 2001). This has led to a convention of using bars to depict values from different nominal categories and using lines to depict values from different places on a metric continuum. This mapping of nominal data to bars and metric data to lines is commonly described in textbooks and guides on graphing—however, there is more to the story. What matters more than the class of the data is the intended message to be conveyed by the visualization. Suppose one wants viewers to compare two values along a continuous dimension—for example, mood at 5 p.m. versus 9 a.m. Using bars to represent mood at those two time points would encourage interpretations such as “Mood is more positive at 5 p.m. than 9 a.m.”; this could be the right choice despite the fact that time is a metric variable. Using the wrong visualization for a given conceptual message can lead to odd conclusions: In the height examples at the bottom right Figure 18, the graphs violate both the data-type convention and the intended conceptual convention. Showing values as a function of weight using a bar graph tends to lead people to conclude that “150-pounders are taller than 125-pounders,” which is reasonable but perhaps not as helpful an insight as “Height increases with increasing weight.” Likewise, showing values as a function of nationality using a line graph could lead a substantial number of viewers to conclude that “One tends to become taller as one becomes more Dutch” (Zacks & Tversky, 1999).

Connect relationships in visualizations to those in the world

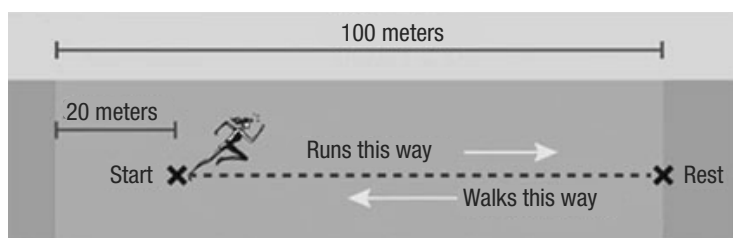
Even if a viewer can extract values and data relationships from a visualization, they can fail to connect those to important real-world concepts in ways that are critical for reasoning. For example, consider the graph and corresponding text at the top of Figure 19, given to 673 high school students by Farrar (2012, as cited in Whitacre & Saul, 2016). Only 4% of the students noted that the text did not correspond to the graph's information. In another study, high school students in honors chemistry and environmental science received similar materials on the topic of teen pregnancy, and only 25% noted a discrepancy between the text and graph (Whitacre &

Changing Global Climate

A friend has asked you to peer edit a presentation. Are all of the facts being presented correctly?

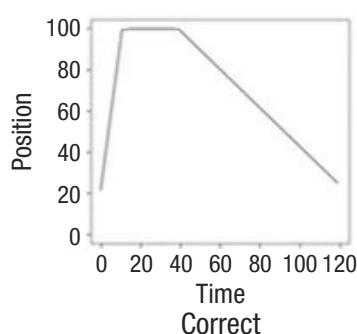
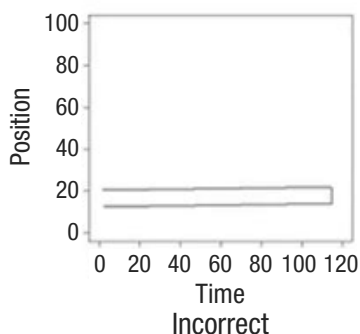


Carbon dioxide (CO_2) emissions data from ice cores was compared with temperature records, confirming that global temperature and carbon dioxide levels are not related.



Gizelle trains on a field that is 100 m long.

She sprints from a point 20 m distant from one end of the field to the other end of the field, takes a short rest, and walks back slowly to the start.



Draw a graph to represent this routine.

Fig. 19. Graphical reasoning problems. When shown the graph at the top, only 4% of high school students noticed inconsistencies with the accompanying text (Farrar, 2012). The bottom shows an example item from Lai et al. (2016); asked to plot a runner's routine on a time–distance graph, most middle-school students drew “maps,” and only 1% drew a correct graph. Bottom figure adapted by permission from Springer: *Journal of Science Education and Technology*, “Measuring Graph Comprehension, Critique, and Construction in Science,” by K. Lai, J. Cabrera, J. M. Vitale, J. Madhok, R. Tinker, and M. C. Linn (<https://doi.org/10.1007/s10956-016-9621-9>). Copyright 2016.

Saul, 2016). In a more complex task (Fig. 19, bottom), sixth through eighth graders were asked to create a position–time graph on the basis of a description of a runner’s training routine. Only 1% of the students did so correctly, and most drew “graphs” representing a map of the runner’s position.

Although students can easily extract individual values and relationships from the climate and running visualizations shown in Figure 19, they have difficulty linking the depicted patterns with the accompanying text because graphing is often taught with a focus on plotting data points and representing quantitative functions. Instead, solving the example problems in Figure 19 requires that they see the underlying link between the visual and verbal representations of the patterns. This problem arises when visualizations must be linked to text, tables, or mathematical equations. One intriguing proposal for helping students see past extracting values, to focus on making external links, is to omit numbers on axes entirely and provide only “qualitative” graphs that distill critical categorical relations. In one study, activities in which students constructed or critiqued qualitative graphs distinctly contributed to their understanding of cancer cell division and treatment impact (Matuk et al., 2019).

In the classroom, another effective method for helping students see these links is to give them abundant and explicit practice in translating between such representations. One study found that prealgebra students generated better solutions for word problems involving simple linear functions when they first generated graphical, tabular, and equation representations of those problems, compared with a control group who generated and solved equations for similar word problems (Brenner et al., 1997). The intervention group also gained a substantially better conceptual understanding of functions compared with the control group. Graphing calculators can also help students quickly see how changing features of equations affects visualized functions (Doerr & Zangor, 2000; Hollar & Norwood, 1999; Mesa, 2008); a meta-analysis of 42 studies of graphing-calculator use found benefits from middle-school mathematics through first-semester college calculus (Ellington, 2006). Another useful technique was to switch rapidly (within seconds) among multiple representations of similar concepts (tables, graphs, equations, and verbal rules), which helped students develop an understanding of functions (Kalchman & Koedinger, 2005).

Another way to facilitate these links is by starting with familiar contexts, such as earning money for each mile walked in a walkathon, allowing students to see abstractions of those functions from concrete meaningful situations (Kalchman & Koedinger, 2005). Likewise,

instructors can provide students with scenarios in which they collect and summarize data and guide them toward inventing visualizations, so that the mappings between the statistical concepts and visualizations are clear (Lehrer & English, 2018; Lehrer & Romberg, 1996; Lehrer et al., 2000).

Visually Communicating Uncertainty and Risk

Why communicating uncertainty is critical

Quantifying uncertainty is critical for analysts and scientists who want to make informed decisions from data. Uncertainty is inherent in the generation of data and models, and it comes from multiple sources. Individual measurements contain noise. Measurements of a population of values typically rely on a sample of those values that is incomplete, or even biased. To estimate average human height, for example, we must rely on measuring a subset of all people. Averaging 100 people’s heights produces an estimate, but an uncertain one, because we did not definitively measure the height of every person in the world. If we assume that the 100 people are randomly sampled from the population, then our uncertainty can be calculated using statistical theory. However, if we were to sample 100 people who all happen to be NBA players, our sample would not be representative of the world, and standard statistical approaches for random samples would not adequately capture the true uncertainty.

Effective communication of uncertainty is necessary for scientific literacy among the public; without understanding uncertainty, people cannot accurately calibrate their internal sense of how reliable a pattern or claim is. Yet communicating uncertainty is challenging as a result of many well-documented biases related to decision making, many of which are characterized by misuse of statistical evidence in reasoning (e.g., Tversky & Kahneman, 1974). Unfortunately, the level of uncertainty in analysts’ calculations is often judged to be too complex or time-consuming to communicate to lay readers or busy decision makers (Fischhoff & Davis, 2014).

Communicating uncertainty is also important for maintaining public trust. In weather forecasts, when uncertainty is presented counterintuitively (or not presented at all), people tend to discount it. Unfortunately, when the forecast turns out to be “wrong,” this can lead to a lack of trust in scientific forecasting (Binder et al., 2016; Joslyn & LeClerc, 2012). Among scientists, underemphasis or misunderstandings of sampling error (a form of uncertainty) and the likelihood of replicating

experimental results may contribute to continued use of underpowered studies and the “replication crisis” across many previously accepted findings (Ioannidis, 2005).

Some common cognitive biases are related to misunderstandings of uncertainty. One reason this may be so is that many core statistical concepts, such as variability, estimation, and sampling, are complex and can be defined clearly only with reference to statistical theory. A key challenge in communicating uncertainty is that lay audiences may not appreciate that estimates are subject to variability and may not be familiar with the statistical abstractions commonly used to express these concepts (Gal, 2002).

This lack of understanding of statistical constructs is present even among researchers. In a *frequentist* statistical paradigm, it is not possible to make statements about the probability that a specific interval contains the true population parameter value (e.g., the true average height of U.S. males) because probability can be defined only as frequency over a long run of trials. Instead, the frequentist 95% confidence interval refers to the probability that a confidence interval constructed in the same way as the plotted interval contains the population parameter. However, even users with statistical training, such as graduate students, are prone to mistakenly conclude that it is 95% probable that a 95% confidence interval contains the true parameter (Hoekstra et al., 2012). Likewise, the relationship between statistical significance and whether or not two error bars overlap is often misunderstood: When two frequentist 95%-confidence-interval error bars do not overlap, it is correct to assume that there is a significant difference between the two quantities at an alpha level of 0.05. However, when the two intervals do overlap, researchers incorrectly assume that there is no significant difference between the two quantities (Belia et al., 2005).

Errors such as these likely stem from challenges that students of statistics and others face concerning how to interpret the sampling distribution of the mean implied by a 95% confidence interval or standard-error interval (Chance et al., 2004; Hullman et al., 2017). The sampling distribution is the distribution of means expected if one were to repeatedly draw samples of a given size n for a population. Although the sampling distribution is a more natural choice when trying to answer questions about the probability that a difference is not zero (i.e., statistical significance) and is systematically related to the variance in the underlying measurements (and in estimates of the parameter value in the population), this distribution does not directly address the sorts of questions one might ask about the effect of applying a treatment in the world.

Common visualizations of uncertainty are often misinterpreted

Uncertainty is challenging to communicate because the concept of uncertainty is difficult to understand in the first place. In graphs designed for a lay audience, depictions of uncertainty are often simply omitted, even when the graphs are intended to support inferences beyond the data sample that is shown (Hullman, 2019). When uncertainty *is* presented, it is most frequently shown through error bars. They typically represent either a standard-deviation range, a standard-error range, or a confidence interval (e.g., a frequentist 95% confidence interval or a Bayesian 95% credible interval). Which of these constructs is depicted has a large influence on a viewer's impression of effect size—referring to a set of quantitative measures of the magnitude of a difference, such as the difference in means between two distributions, divided by their pooled standard deviation (Hofman et al., 2020). For example, when viewing results of an evaluation of a new intervention (e.g., a special type of equipment one might use to improve one's score in a sports competition) relative to a control, one might wonder how much investing in the new equipment is likely to help increase one's score for a randomly drawn instance of the game. When shown frequentist 95% confidence intervals representing uncertainty on bars representing the control (“Standard Boulder”) and treatment (“Special Boulder”) outcomes in a fictional boulder-sliding competition (Hofman et al., 2020; see Fig. 20, left), lay viewers were inclined to pay more for the treatment and overestimate effect size relative to when they saw error bars showing a predictive interval representing uncertainty in the underlying measurements (Fig. 20, middle). Although not as effective as showing predictive intervals, rescaling the y -axis of a chart showing 95%-confidence-interval error bars to display the range required for the predictive interval slightly reduced overpayment and bias in effect-size estimations (Fig. 20, right).

Error bars can lead lay participants to use visual reasoning strategies that result in error-prone perceptions of effect size. For example, when lay participants were asked to evaluate the respective effect size between two distributions shown as bars representing means with error bars, their answers appeared to differ mostly on the basis of the difference between the means and were unaffected by the level of uncertainty around those means (Hullman et al., 2015). One investigation of the visual reasoning strategies employed by lay participants to make effect-size judgments and decisions from error bars and other common uncertainty visualizations found that many participants used approximations based on visual distances in the chart

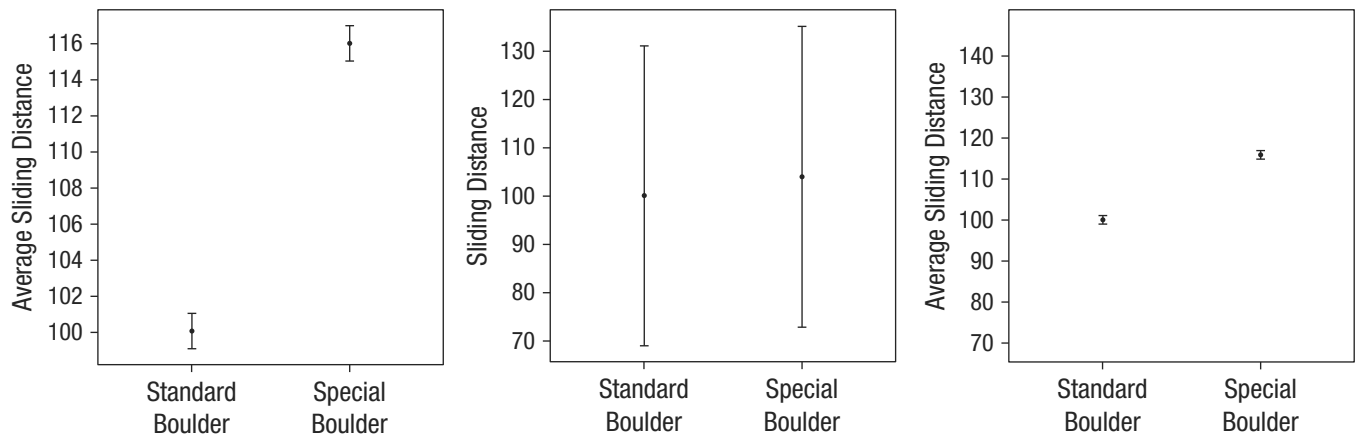


Fig. 20. Examples of misunderstandings of variance depictions. Hofman et al. (2020) found that participants in an online study who viewed the chart on the left, showing a range extending $1.96 SE$ above and below the mean, were more likely to pay more for a new intervention (i.e., “Special Boulder”) compared with the standard intervention (i.e., “Standard Boulder”) and to overestimate the size of the new intervention’s effect than those who saw the chart in the center, showing a range extending $1.96 SD$ above and below the mean. The chart on the right is the same as the chart on the left, except that it has the y -axis range of the chart in the center. This display slightly reduced overpayment and overestimation, which suggests that viewers are sensitive to superficial visual cues, such as how big a difference looks within the frame of the chart. Republished with permission of the Association for Computing Machinery, from “How Visualizing Inferential Uncertainty Can Mislead Readers About Treatment Effects in Scientific Results,” by J. M. Hofman, D. G. Goldstein, and J. Hullman, *CHI ’20: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (<https://doi.org/10.1145/3313831.3376454>). Copyright 2020 Association for Computing Machinery.

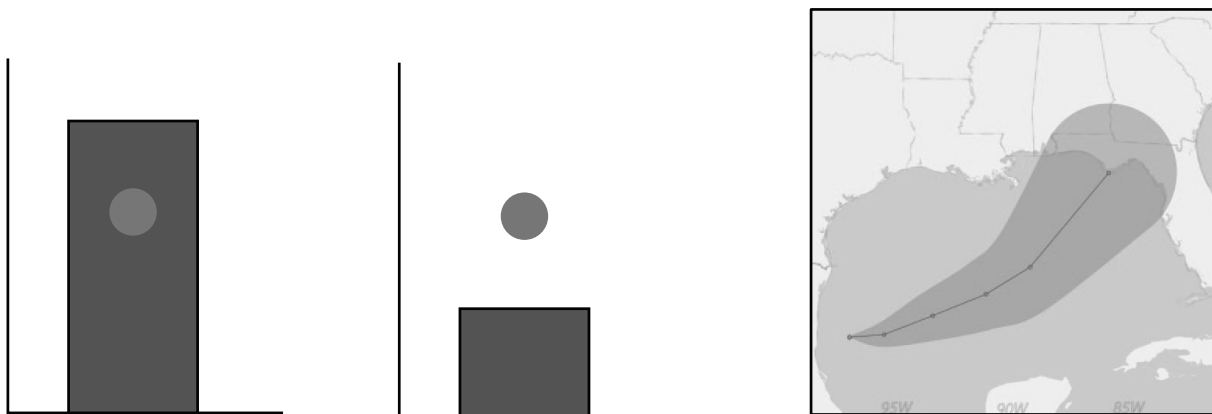


Fig. 21. Examples of misunderstandings of the boundaries of graphical objects. At left, the dots below and above the bars are equidistant from the bars’ tips, but viewers rate the point inside the bar as being more likely to belong within the distribution the bar summarizes. At right, the graph presents an adapted version of the cone of uncertainty created by Le Liu and Donald House and tested in Ruginski et al. (2016).

rather than reasoning more deliberately about the size and uncertainty of effects (Kale et al., 2021).

If lay viewers are unfamiliar with standard-deviation and confidence intervals, it is perhaps unsurprising that they tend to misunderstand the depictions of those parameters through error bars or confidence envelopes. Error bars are not seen as parameter estimates for a continuous distribution but are instead mistaken as depicting the discrete range of the data. When shown bars representing 95% confidence intervals of a forecasted nighttime low temperature, participants incorrectly believed that the upper and lower bounds represented forecasted high and low temperatures

(Joslyn & LeClerc, 2012). This mistake occurred despite the presence of a highly visible key that depicted the correct interpretation. Figure 21 illustrates how not just error bars but bar graphs themselves can lead to similar misunderstandings of uncertainty: Viewers appear to implicitly believe that a bar contains the full distribution of data values that it summarizes because of its metaphorical status as a container. When reminded that the tip of a bar graph represents the mean of a distribution of values, viewers still rate points that fall slightly below the tip as more likely to belong to that distribution than points placed slightly above it (Correll & Gleicher, 2014b; Newman & Scholl, 2012).

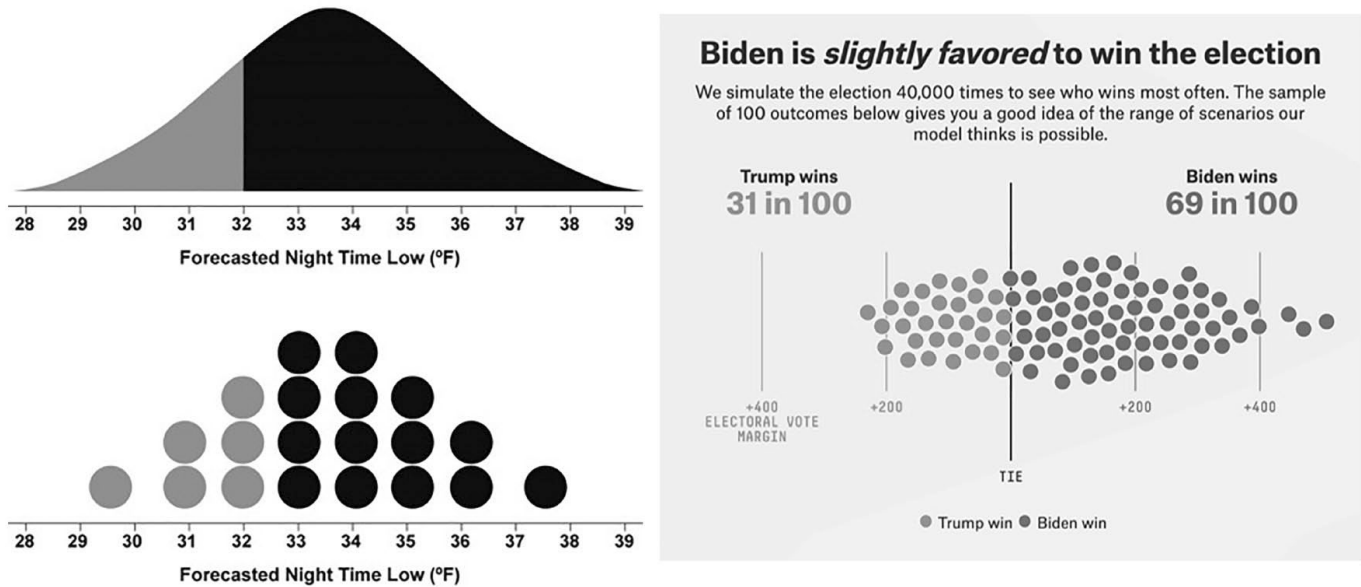


Fig. 22. Depictions of distributions. At left, a probability density plot is represented as a quantile dot plot that uses a frequency (discrete-outcome) framing. Here, black dots or shaded regions indicate values above freezing temperatures. At right, a jittered variation with a categorical variable added via color shows uncertainty surrounding the 2020 U.S. presidential election outcome. Graph at right adapted by Anna Wiederkehr of FiveThirtyEight (for more on FiveThirtyEight’s creation of similar graphics, see Wiederkehr, 2020). Used with permission from FiveThirtyEight.

Another notable example of this “discrete range” error occurs with hurricane-forecast visualizations. The most popular method for displaying hurricane paths, produced by the National Hurricane Center (see Fig. 21), uses a “cone of uncertainty” to represent 66% confidence intervals across thousands of storm-track simulations. Note that this means that many predicted storms would travel outside of this cone. Unfortunately, people incorrectly believe that the cone depicts a discrete range, such that areas in the cone are in danger and areas outside are relatively safe (e.g., Padilla et al., 2017, 2020). Even worse, many viewers do not recognize the cone as reflecting the storm’s potential paths, instead assuming that it depicts growth in the storm’s size over time.

How to visualize uncertainty more intuitively

Instead of depicting summary statistics (e.g., confidence intervals) that are rarely well understood by lay audiences, a better approach is to show the underlying probability distribution. Density plots (Fig. 22, top left) and violin plots (similar, but typically mirrored to show a single symmetric shape) show distributions by mapping the probability of a given value to width. These plots can help viewers intuitively understand the predicted variability of values, location (e.g., mean), and distribution shape (e.g., normal, skewed). They can also promote intuitive assessments of how likely or

surprising different values are, in ways that are more closely aligned with normative statistical definitions (Correll & Gleicher, 2014b). Several studies have found that density plots lead to better-quality decisions than error bars showing predictive or confidence intervals (Fernandes et al., 2018; Hofman et al., 2020; Kale et al., 2021).

In other work, some researchers have argued that visual features such as fuzziness, color value, and unordered line arrangement convey probability more naturally than more abstract mappings, such as mappings between uncertainty and size (MacEachren et al., 2012; see Fig. 23, left). However, representations that were judged to be more associated with uncertainty did not necessarily lead to faster or more accurate judgments. These findings illustrate a trade-off whereby some encoding types provide a subjective impression of uncertainty but may also reduce precision, making some tasks hard to complete. Experts might therefore prefer more precise representations because they have no trouble understanding how these visualizations convey uncertainty, whereas lay viewers likely benefit more from less precise visualizations that are more semantically resonant (or more naturally interpreted as uncertainty).

Yet another approach for showing statistical uncertainty intuitively is to rely on visual depictions that are inherently *perceptually* uncertain, preventing the viewer from resolving a value precisely. If the location of a point on a map could be blurred proportionally to the uncertainty in the position, then the recovered position

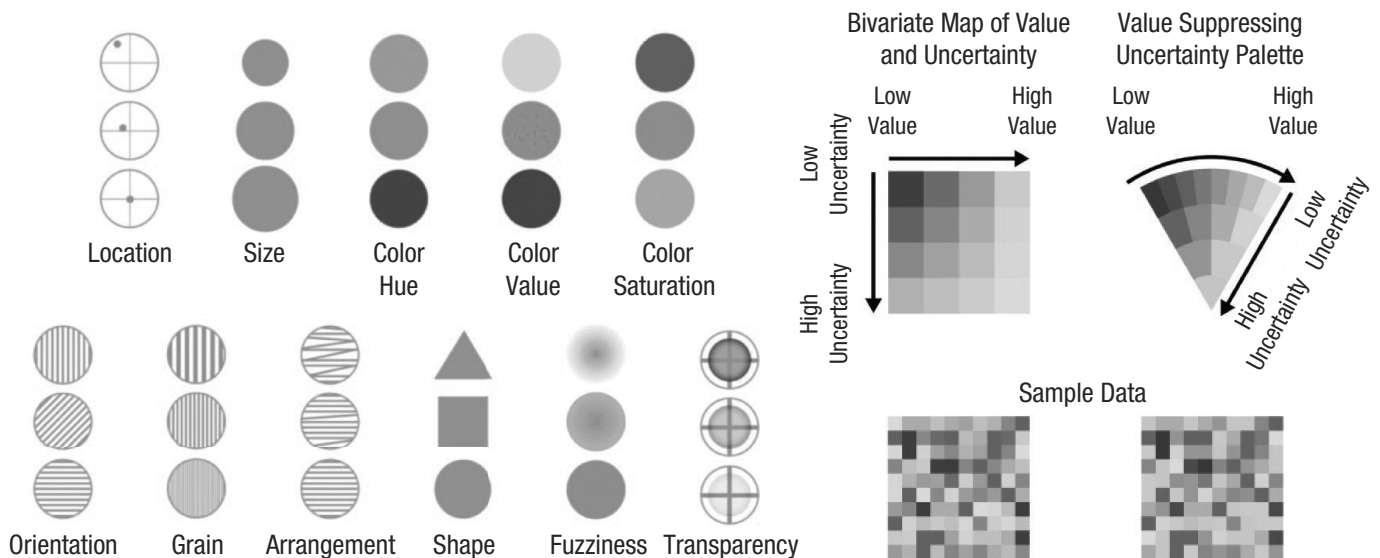


Fig. 23. Methods for depicting uncertainty. On the left of the figure are different visual variables for conveying uncertainty tested by MacEachren et al. (2012), who found that fuzziness, location, arrangement, and color value were rated as relatively more logical than other variables for representing uncertainty. The right shows a bivariate color map that separately encodes value (via hue) and uncertainty (via lightness). A value-suppressing uncertainty palette uses a tree metaphor to collapse value distinctions in proportion to the uncertainty surrounding the values (Correll et al., 2018). Left figure reprinted with permission from “Visual Semiotics & Uncertainty Visualization: An Empirical Study,” by A. M. MacEachren, R. E. Roth, J. O’Brien, B. Li, D. Swingley, and M. Gahegan, 2012, in *IEEE Transactions on Visualization and Computer Graphics*, 18(12), p. 2497 (<https://doi.org/10.1109/TVCG.2012.279>). Copyright 2012 by IEEE. Right figure republished with permission of the Association for Computing Machinery, “Value-Suppressing Uncertainty Palettes,” by M. Correll, D. Moritz, and J. Heer, *CHI ’18: Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (<https://doi.org/10.1145/3173574.3174216>). Copyright 2018 Association for Computing Machinery.

would intuitively contain at least that level of uncertainty. Value-suppressing uncertainty palettes attempt this for color, taking advantage of a natural relationship among hue, saturation, and lightness: Hues are easier to tell apart when they are more saturated and darker (Correll et al., 2018; Fig. 23, right). Value is mapped to hue and certainty to saturation and lightness (darker for more certain). As a result, value judgments are more difficult for more uncertain values—the most uncertain values all appear as the same shade of gray. When this approach is applied to maps, users weigh uncertainty more heavily than they do when using conventional bivariate maps separating value and uncertainty.

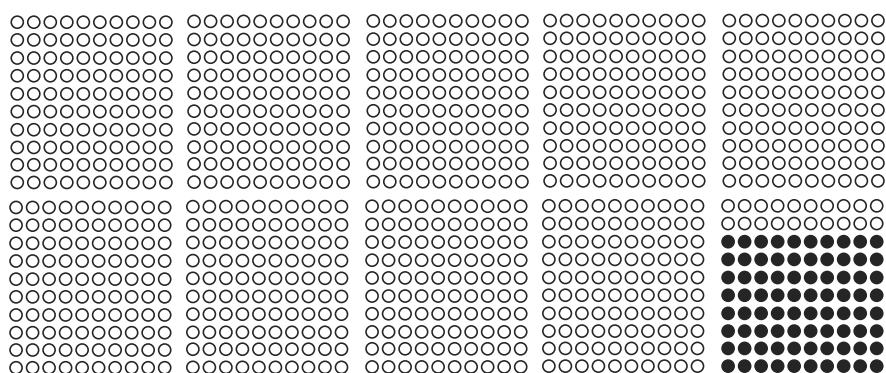
Present uncertainty as examples over space or time. Mounting evidence suggests that individuals can more effectively understand probabilities (e.g., 10%) when they are communicated as frequencies (1 out of 10; e.g., Hoffrage & Gigerenzer, 1998; Peters et al., 2011; Schapira et al., 2001). For example, individuals were informed about a headache medication’s side effects, framed as either a percentage (e.g., 10% of patients got a bad rash from the medication) or a frequency (e.g., 10 out of 100 patients got a bad rash from the medication),

before they rated the riskiness of the treatment. Individuals with lower numeracy rated the treatment as less risky when the information was communicated with a percentage, as opposed to a frequency, framing. Individuals with high numeracy did not show a difference in their riskiness ratings as a function of framing. These results imply that the framing influenced individuals with low numeracy because the percentage was harder for them to understand than the frequency, whereas those with high numeracy understood both formats equally well (Peters et al., 2011). The frequency framing may be more effective because it more naturally maps to how we experience probability in our daily lives (Hoffrage & Gigerenzer, 1998).

Convey probability with frequency-based visualizations. Frequency-based visualizations are those that express probability as frequency or ratio, as in a set of icons that show the numerator and denominator (icon arrays; see Fig. 24) or dots that represent discrete probabilities in a distribution (quantile dot plots; see Fig. 22, left). The defining characteristic of frequency-framing visualizations is that they allow the viewer to infer probability via the frequency of visual attributes (e.g., position or size) of discrete visual elements such as icons, dots, or lines.

For people with symptoms of arterial disease, aspirin can reduce the risk of having a stroke or heart attack by 13%.

Without aspirin



With aspirin

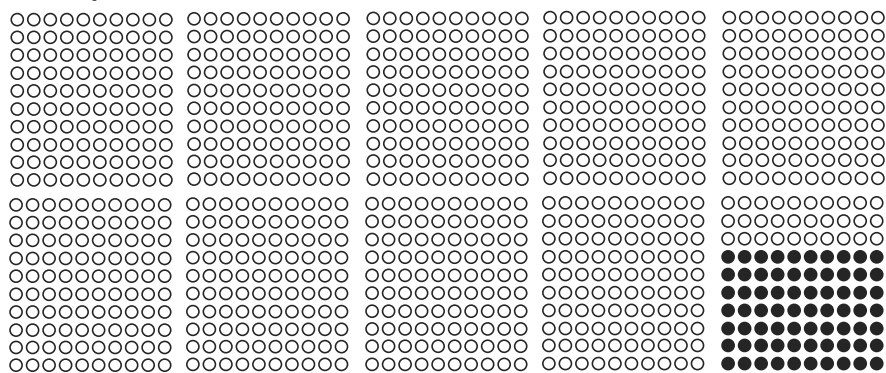


Fig. 24 Icon arrays from Galesic et al. (2009), which help viewers visually compare the relative rate of stroke or heart attack with and without aspirin. Copyright © 2009 by the American Psychological Association. Adapted with permission from “Using Icon Arrays to Communicate Medical Risks: Overcoming Low Numeracy,” by M. Galesic, R. Garcia-Retamero, and G. Gigerenzer, *Health Psychology*, 28(2), p. 211 (<https://doi.org/10.1037/a0014474>).

For example, a substantial body of research has demonstrated that icon arrays are one of the most effective ways to communicate probability in a health-care context because they use frequency framing (Fagerlin et al., 2005; Feldman-Stewart et al., 2007; Garcia-Retamero & Galesic, 2009; Garcia-Retamero et al., 2010; Hawley et al., 2008; Tait et al., 2010; Waters et al., 2006, 2016). In one study, students and older adults read descriptions of multiple medical scenarios framing the effects of treatments in terms of either absolute or relative health risks (e.g., “For people with symptoms of arterial disease, aspirin can reduce the risk of having a stroke or heart attack by 13%” vs. “8% of such people who did not take aspirin had a stroke or heart attack, compared with 7% of such people who did take aspirin”; Galesic et al., 2009). For a second group of participants, each text description was accompanied by two icon arrays

that reinforced the numeric information (see Fig. 24). When asked to estimate outcomes with and without treatment (e.g., how many people out of 1,000 would have a stroke or heart attack if they did and did not take aspirin), participants who had received the additional icon arrays were significantly more accurate than those who had received only text-based probabilities. The benefits of icon arrays were observed for both students and older adults, and individuals with low numeracy drove the effect (Galesic et al., 2009; see also Garcia-Retamero & Galesic, 2009).

Researchers have conceptually replicated the advantages of communicating ratio data as icon arrays numerous times (e.g., Garcia-Retamero et al., 2010; Okan et al., 2012; Stone et al., 2003; Zikmund-Fisher et al., 2014; for a meta-review, see Garcia-Retamero & Cokely, 2017). Another study also found that patients trust icon

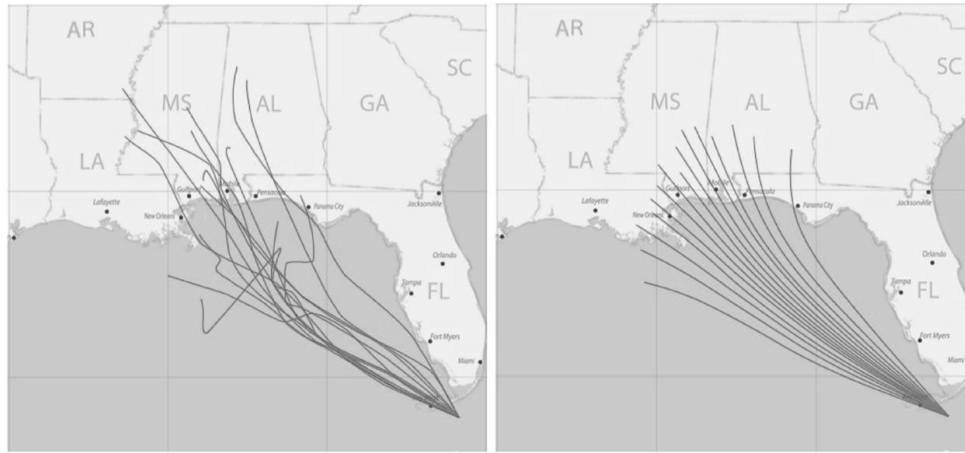


Fig. 25. Techniques for depicting uncertainty in hurricane paths. At left, randomly sampled ensemble hurricane paths, known as spaghetti plots, can look messy. At right, a new procedure for reconstructing ensemble paths can show the distribution's full spread and reduce visual clutter. When paths are easy to distinguish, additional information—such as the storm's size and intensity—can be added to the plot (L. Liu et al., 2018).

arrays more than other common visualization techniques (Hawley et al., 2008), and other work has confirmed that icon arrays help patients with low numeracy correctly interpret probabilities (e.g., Galesic et al., 2009; Garcia-Retamero & Galesic, 2009; Hawley et al., 2008).

A frequency-framing approach is also recommended for conveying distributional information. In the hurricane cone-of-uncertainty example in Figure 21, viewers apply the wrong schema when translating the visualization back to the data that it represents. Viewers misinterpret the edge of the cone, which depicts error bars showing 66% confidence intervals, as ranges (Ruginski et al., 2016) or mismatch the width of the cone to the storm's size instead of its potential path (Padilla et al., 2017). Given that policymakers and the general public rely on hurricane-path information to make decisions about preparation and potential evacuation, it is critical that they understand the depicted information immediately and intuitively.

Because it seems unrealistic to attempt to teach lay viewers both the requisite underlying statistical concepts and the schema for how they are mapped to a visualization, a more fruitful route is to redesign the visualization to use a frequency-based approach that pulls from existing metaphors known to lay viewers. One strategy is to simply show numerous examples of the samples that created the summary statistics (or simulated draws from the population estimate). This mimics the experience of collecting samples in the real world: flipping coins to gauge the percentage of tails, trying multiple menu items to gauge a restaurant, or testing how various kicking styles might put spin on a

soccer ball. Providing discrete-outcome framings of data can provide a frequency metaphor that reduces probability-processing errors (Hullman et al., 2015; Kale et al., 2019, 2021; Kay et al., 2016; Padilla et al., 2017).

To apply this alternative approach to hurricane forecasts, one solution to misunderstandings of the cone of uncertainty is to instead plot a sample of possible hurricane paths, allowing the viewer to intuit which areas are in most danger (Fig. 25). This method stops viewers from confusing uncertainty about the hurricane's path with the size of the storm and helps them understand which areas are most endangered (L. Liu et al., 2018). When showing uncertainty information using a frequency metaphor, one must ensure that the samples shown are representative of the uncertainty in the data. In hurricane forecasting, track visualizations are sometimes referred to as "spaghetti plots" because they can look messy, like a plate of noodles, when randomly sampled (Fig. 25, left). To increase the usefulness of this technique, researchers have developed reconstruction procedures to ensure that the tracks clearly show the full uncertainty in a storm's trajectory (see Fig. 25, right; L. Liu et al., 2018).

Density and violin plots are useful for conveying the data's shape, but it can be difficult to use them to extract specific *cumulative* values (What is the probability of a randomly selected value's being less than this one?), which requires the viewer to visually calculate an integral. In contrast, quantile dot plots provide a generalizable approach for showing a distribution yet also make values easier to read, by providing probabilities not only through the heights of the function (as in the density plot) but also through the (easily countable)

number of dots that contribute to that height. As seen in Figure 22 (left), a quantile dot plot represents a distribution using stacked dots that account for a specific portion of the data; in this case, each dot depicts a 5% probability.

Using the graph at the left of Figure 22 as an illustration, imagine that the viewer's task is to decide whether the nighttime low temperature will drop below freezing. With the quantile dot plot, the viewer could count the dots to determine that there is a 30% chance the temperature will be 32°F or lower. This same task is much harder with a density plot because it requires the viewer to visually calculate an integral under the curve. Evidence suggests that quantile dot plots improve accuracy and memory compared with density plots (Hullman et al., 2017; Kay et al., 2016) and outperform summary plots, density plots, and text descriptions of uncertainty for decisions with risk (Fernandes et al., 2018).

To present their 2020 U.S. presidential forecast, FiveThirtyEight used a variant of a quantile dot plot, which they called a “ball swarm” plot, showing 100 random samples from their model (Fig. 22, right; for details on the creation of these plots, see Wiederkehr, 2020). Traditionally, dot plots have been used to show distributions of experimental samples (Wilkinson, 1999). In the ball-swarm plot used by FiveThirtyEight, a total of 100 balls representing 100 hypothetical elections were clustered along an x -axis denoting the margin by which the election could be won by either candidate. Some balls were colored red in proportion to the forecast's current predictions of Donald Trump's chance of winning; the remainder were colored blue to depict Joe Biden's chances.

Showing multiple samples or simulated draws from a distribution can more intuitively communicate that distribution to viewers without training in statistics. Showing those samples simultaneously can lead in many cases to accurate estimates of probability, but in other cases it can lead to new misconceptions. For example, when shown a sample of possible hurricane tracks, viewers often see that set as reflecting the entire population of possibilities (no more and no fewer) instead of merely a subset. When shown visualizations such as those in Figure 25, viewers can overreact if one of the paths directly hits their town (Padilla et al., 2020), even though the locations between neighboring paths are generally as likely to be hit as the locations underneath a path.

One solution to this problem is to show those samples serially across time instead of simultaneously in space, relying on the limited capacity of working memory to force viewers to naturally abstract those samples into intuitive statistics. Hypothetical outcome plots (HOPs; Hullman et al., 2015; Kale et al., 2019) animate a set of random draws from a distribution, showing

each draw for a short duration (< 500 ms). Figure 26 presents an example set of frames in which each frame is one random draw from the joint distribution represented by the error bars on the left. The frames are shown in a random sequence, which creates an animation that can give viewers an intuitive sense of the uncertainty surrounding the true trend.

Presenting animated draws over time has several advantages relative to static approaches. First, joint probabilities (e.g., What is the chance that the rate of disease really is higher in Location A than in Location B?) are naturally expressed as patterns over time, but showing them in a static visualization would require intentionally choosing an encoding capable of presenting joint probability (e.g., a heat map). The visual system's natural ability to extract summary statistics from temporal frequency (Hasher & Zacks, 1984) can allow viewers to judge effect size more easily than they could with static depictions of each distribution (e.g., as bars with error bars or violin plots, which do not convey dependence between distributions).

Second, when viewers may be likely to ignore uncertainty in favor of simpler heuristics—for instance, judging an effect size using only the size of the visual difference between the means of two variables, regardless of axis scaling—any static visualization that encodes central tendency runs the risk of allowing users to discount uncertainty in their judgments. In such cases, viewers may ignore uncertainty or use an overly simple strategy, whether a mean is depicted using a direct summary mark, such as a mean temperature forecast presented along with a confidence interval of the mean (Joslyn & LeClerc, 2012), or implicitly, on the basis of the highest (or widest) point in a density or violin plot of a Gaussian distribution (Hullman et al., 2015; Kale et al., 2021).

Finally, in visualizations that use the most precise visual channels to show data—for example, a map that uses both vertical and horizontal dimensions of position to show geographic locations—it can be difficult to convey uncertainty because that requires adding a new visual channel (e.g., saturation of position markers) to an already complex visualization. However, as long as a visualization is not already animated, HOPs can be used without requiring the designer to choose a new way to depict uncertainty. This has inspired visualization researchers to use probabilistic animation to show uncertainty in depictions of geospatial data (Ehlschlaeger et al., 1997; Fisher, 1993) as well as other complex visualizations (Feng et al., 2010).

Designers might also draw from the best of both worlds by combining animated draws with static displays. Each animated draw of a hypothetical outcome plot could leave a trace that slowly builds into a static

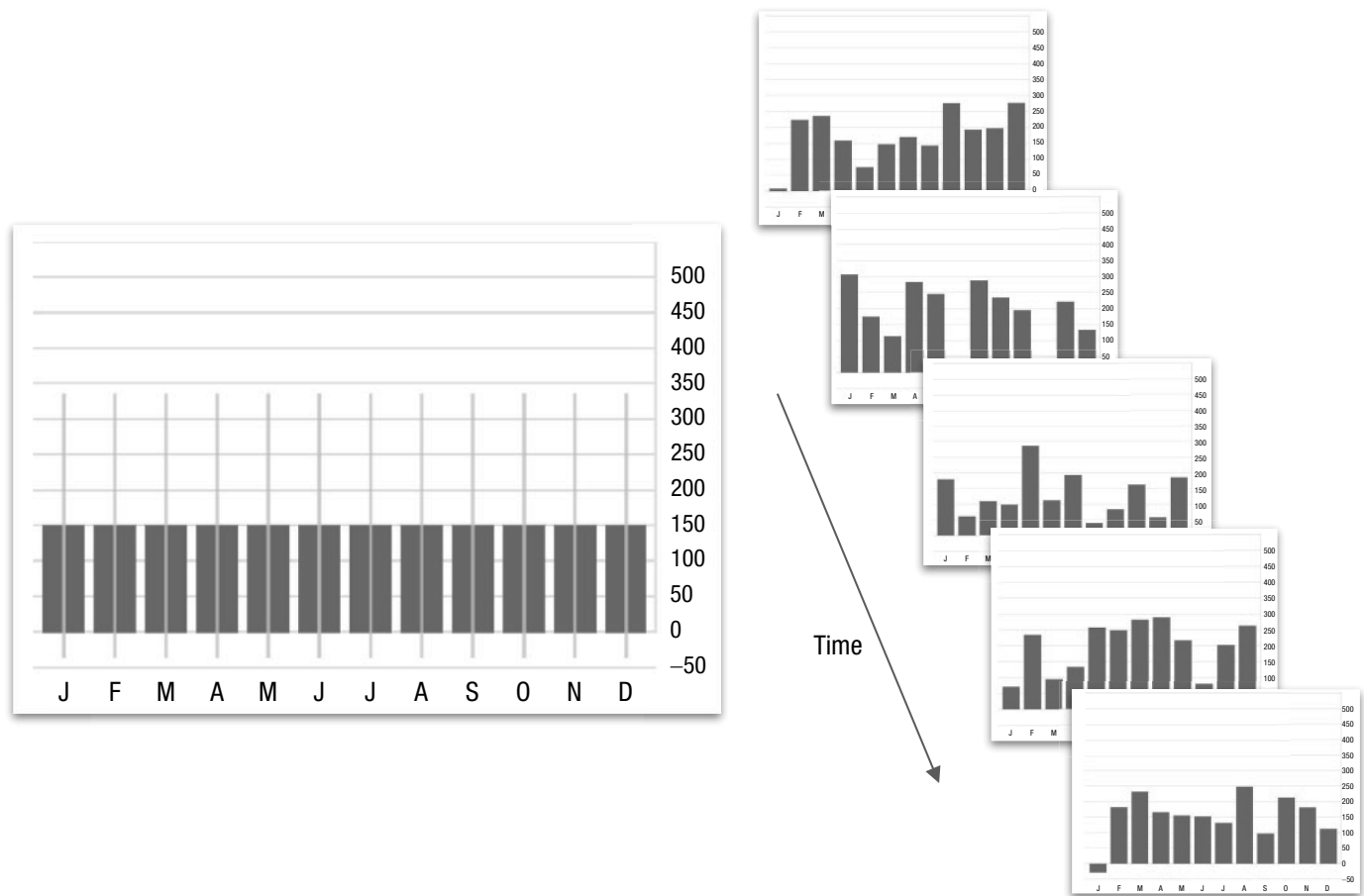


Fig. 26. Example of a hypothetical outcome plot. An example set of frames (right) in which each frame is one random draw from the distribution on the left. The frames are shown in a random sequence, which creates an animation that can give viewers an intuitive sense of the uncertainty in the true mean (Hullman et al., 2015).

display such as a gradient plot, or animated draws could help visually explain a static technique such as a density plot or error bar. *The New York Times* presented animated dots in a simulation to show inequalities in wealth distribution due to race (Badger et al., 2018). This combination of animated and static depictions of uncertainty is fertile ground for research at the intersection of psychology, statistical cognition, and data visualization.

How to communicate health risk

The rapid advancement of technology has made it increasingly easy to collect and analyze massive amounts of health-related data (Ola & Sedig, 2016). In addition to traditional medical records, doctors and patients now have access to information from population surveys, genomic sequencing, ancestral records, wearable devices, and medical implants. Given the high volume, rapid development, and diversity of this information, it is understandable that making decisions with

health data can be complicated. A large body of research is dedicated to examining if visualizations can help doctors and patients incorporate data into health and wellness planning (for reviews, see Ancker et al., 2006; Garcia-Retamero & Cokely, 2017; Garcia-Retamero et al., 2012; Lipkus, 2007; Lipkus & Hollands, 1999).

A consistent finding from research on the visual communication of health data is that visual aids can clarify numeric information for both patients and doctors. A systematic review of visual aids in health research found that effectively designed visual aids can improve risk assessments, resulting in reductions in health-decision biases, improvements in healthy behaviors, and increased trust in treatments (Garcia-Retamero & Cokely, 2017). It is noteworthy that the benefits of visual aids are strongest for individuals with low numeracy (Lipkus et al., 2001) and low risk literacy (i.e., ability to evaluate one's risk; Ancker & Kaufman, 2007). Low numeracy tends to correlate with less advanced education, which disproportionately affects minoritized groups (Rodríguez et al., 2013)—however, one study

with 463 college-educated participants found that 16% to 20% incorrectly answered relative simple questions about risk and probability (e.g., “Which represents the larger risk: 1%, 5%, or 10%?”; Lipkus et al., 2001).

For visual aids to effectively help people reason about health-care data, visualizations must be designed with care (Fagerlin et al., 2011). The following section provides a summary of what works in the communication of health-care data, drawing from the most consistent and reliable evidence-based findings to date.

Convey risks with absolute—not relative—rates. A rational decision maker should not make different treatment decisions based on the way that medical risk is conveyed. Unfortunately, the presentation of medical-risk information has a profound influence on judgments, even for highly educated people. In a study of 235 practicing physicians who were asked to indicate how likely they were to treat patients for hypertension and hypercholesterolemia on the basis of information about the treatments; 41.3% were more likely to recommend a treatment when its effects were framed as a relative reduction in risk compared with an absolute reduction in risk (Farrow et al., 1992). Absolute risk is the overall probability of experiencing something, such as getting a disease, during some period of time. Relative risk is a comparison between the absolute risk of two groups (Natter & Berry, 2005). As Fagerlin et al. (2011) illustrated, the change in breast cancer risk from 4% to 2% associated with the drug Tamoxifen can be communicated in a relative format (a reduction of 50%) or an absolute format (an absolute reduction of 2%).

Absolute-risk formats are preferable: Many studies have found that the same treatment outcomes are perceived more favorably when communicated in terms of relative risk as opposed to absolute risk (for a meta-analysis, see Covey, 2007). Further, relative-risk framing can make changes in risk appear larger than absolute-risk framing (e.g., a 50% relative change incorrectly seems larger than a 2% absolute change; Akl et al., 2011; Baron, 1997; Farrow et al., 1992; Malenka et al., 1993). This framing bias makes physicians more likely to prescribe interventions when information about their effects is communicated using a relative-risk format (e.g., Bucher et al., 1994; Lacy et al., 2001; Naylor et al., 1992). A large body of evidence suggests that the absolute-risk format leads to the least biased decisions. Although this has not been tested with visualizations, the evidence points to absolute risk as the best choice.

Convey ratios with icon arrays and pictographs. In addition to conferring the previously described benefits of frequency framing, icon arrays help to reduce common errors in health-care communication, such as individuals’ focusing on the numerator and neglecting the

denominator, an effect called denominator neglect (for review, see, Garcia-Retamero et al., 2012). For example, when comparing a cancer with a mortality rate of 1,286 out of 10,000 people to a cancer with a mortality rate of 24.14 out of 100 people, undergraduate participants reported that the former cancer was riskier (Yamagishi, 1997). Researchers have proposed that individuals pay more attention to the relative differences in numerators (in this case, 1,286 vs. 24.14 deaths), even though they should consider the relative ratios (12.86% vs. 24.14% mortality; e.g., Garcia-Retamero et al., 2012; Yamagishi, 1997). A variety of studies (including Galesic et al., 2009, described above) have shown that icon arrays can reduce denominator neglect by allowing patients to compare relative ratios visually (see Fig. 24; see also Garcia-Retamero & Galesic, 2009).

Icon arrays can help patients overcome common biases. In one study, participants read statistical information about two treatment options for angina—bypass surgery, which is relatively onerous but 75% successful, and balloon angioplasty, which is less onerous but only 50% successful—and anecdotes about individuals’ experiences with each procedure that were either unrepresentative or representative (50% positive and 50% negative for both procedures vs. 75% positive and 25% negative for bypass surgery). Some participants were also provided with icon arrays that reinforced the statistical information. Of the participants who did not see the icon arrays, 41% who received more anecdotes about bypass surgery’s positive effects reported that they would likely prefer surgery. Surprisingly, only 20% of those who read fewer positive stories, without the additional icon arrays, responded that they would undergo surgery. When those participants were provided with the icon arrays, the effect of the anecdotes disappeared (Fagerlin et al., 2005).

To avoid denominator neglect, it is best to use icon arrays that clearly show part-to-whole comparisons by putting both the denominator and the numerator in the same array (as in Fig. 24; Garcia-Retamero & Cokely, 2017). Designers should avoid showing only the numerator with icons, and leaving the denominator only in the text, because viewers will be likely to neglect the denominator (Stone et al., 2003). Icons should also be arranged into a systematic grid that makes them easy to count, because icon arrays that are not arranged systematically are challenging to use (Feldman-Stewart et al., 2007), particularly for viewers with low numeracy (Ancker et al., 2011; Zikmund-Fisher et al., 2012).

Designers can also consider using anthropomorphic icons, such as outlines of people. In one study, people saw icon arrays of their personalized cardiovascular risks based on data about their age, weight, and health metrics. Specifically, each individual was shown one of

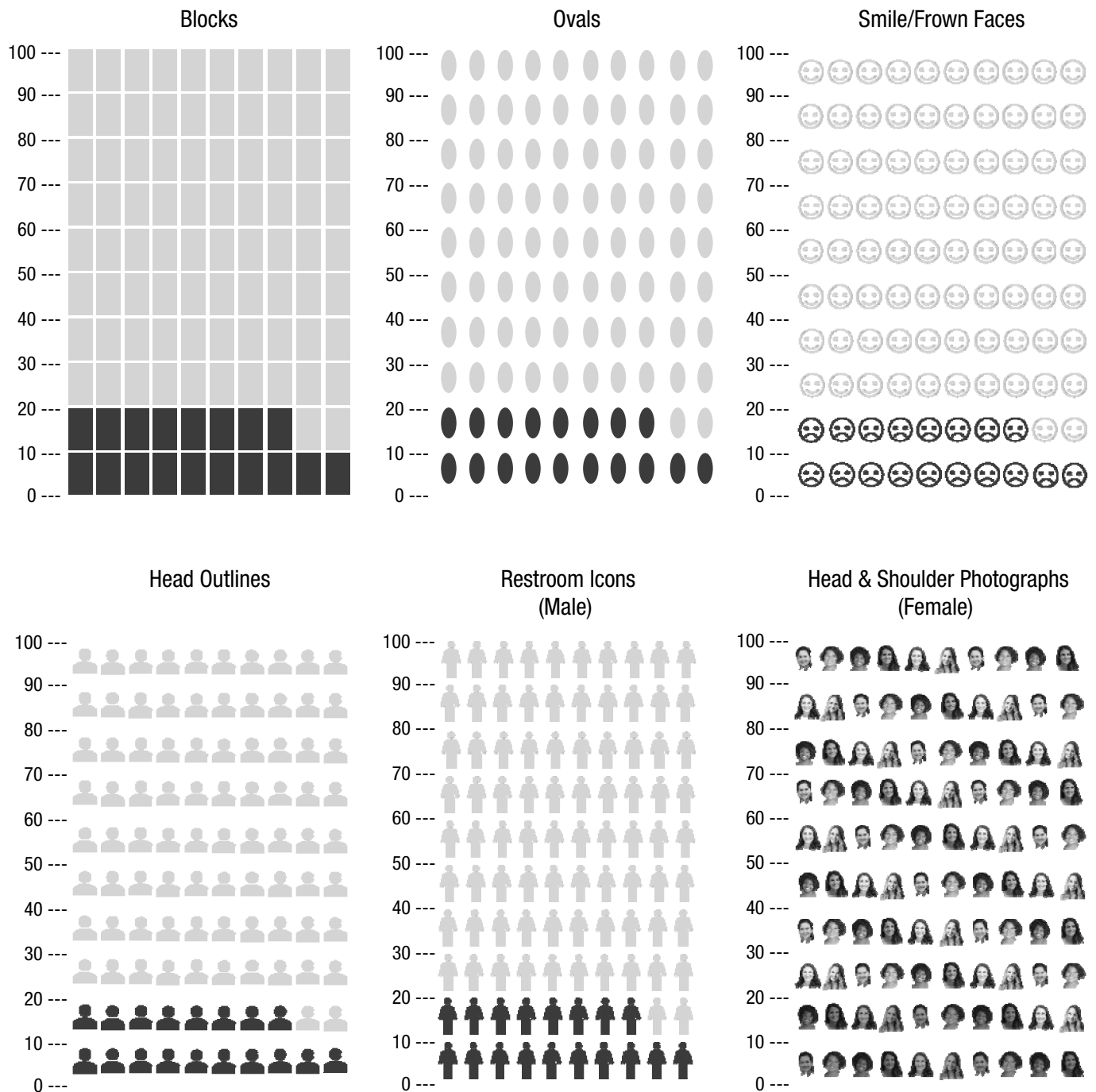


Fig. 27. Icon arrays tested in Zikmund-Fisher et al. (2014). Original figure published under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.

the icon-array formats shown in Figure 27 and was then asked to report their perceived personal risk on a scale from 0 (*extremely small*) to 100 (*extremely big*). People were also asked to report how likely they thought they were to have heart disease or a stroke within 10 years and were later asked how well they remembered the icon array's information. Individuals with higher graph literacy and numeracy who were shown restroom icons

had the highest correlation between their perceived risk and actual risk (Zikmund-Fisher et al., 2014). By contrast, those with low graph literacy and numeracy had the same relationship between their perceived risk and actual risk across all visualizations. In addition, participants' memory for their risk scores was more accurate when they were shown anthropomorphic icons (restroom icons, heads, or photos), particularly for those

with low graph literacy and numeracy. Participants also rated the restroom icons as the most preferable. An anthropomorphic font called Wee People provides an easy-to-use set of icons depicting realistic human silhouettes and is available under a Creative Commons license (Cairo & Klein, 2018).

Summary of Key Guidelines

- A viewer's visual system can extract broad statistics about the data within a display, such as the mean and extrema, within a fraction of a second. Visualize your data with histograms and scatterplots before trusting statistical summaries.
- Beware common visual illusions and confusions. Failing to start axes at zero can cause viewers to overestimate differences. When plotting data with circles or squares, map the data to their areas, not their diameters. The differences between lines in a line graph are increasingly perceptually distorted as the lines increase in slope. Do not plot intensities on intensities, which causes contrast illusions. Mapping a continuous set of numbers to a spectrum of different hues exaggerates differences that happen to straddle the hue boundaries. For accessibility of color-blind viewers, pair red with blue instead of green.
- Although extracting global statistics is fast, comparisons between subsets of values are slow—limited to only a handful per second. So use visual grouping cues to control which set of comparisons a viewer should make, and use annotation and highlighting to narrow that set to the single most important comparison that supports your message. In a live presentation, rely on language and gesture to illustrate what you see. Do this even when you feel it is not needed: Presenters suffer from a “curse of knowledge” that causes them to overestimate how well others see what they see.
- Avoid taxing working memory by converting legends into direct labels. When possible, integrate relevant text into visualizations as direct annotations. Avoid animations, which typically lead to confusion. Graphical embellishments, sometimes derided as “chart junk,” can distract if unrelated to the data, but if they are related, they can improve viewers' memory and engagement.
- New visualization formats must be learned, so try to rely on formats that are familiar to your audience. Respect common associations, such as “up” mapping to “more” for vertical position and “more opaque” mapping to “more” for intensity.
- Graph comprehension depends on both bottom-up and top-down factors. Use bottom-up visual

salience and top-down direct labels to drive attention to relevant features. Use a graph format that guides viewers to the conceptual message you are trying to convey, respecting their previous experience with graphs.

- When communicating uncertainty to a lay audience, avoid error bars, which can be misinterpreted as data ranges. Instead, show examples of discrete outcomes, either simultaneously or over time.
- When communicating risk to low-numeracy audiences, rely on absolute instead of relative rates, convey probabilities with frequencies (e.g., 3 out of 10) instead of percentages (e.g., 30%), and use well-constructed icon arrays with the same denominator.
- Supporting comprehension and understanding is especially important when the intended audience may have low domain knowledge, knowledge about graphing conventions, numeracy, or working memory capacity.

Transparency

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Declaration of Conflicting Interests

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